

BACAU, Romania

WMO: 151500

Lat: 46.5578N

Lon: 26.8967E

Elev: 175

StdP: 99.24

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.3	-12.2	-17.8	0.8	-14.6	-14.6	1.1	-11.5	10.5	-4.3	9.0	-2.3	1.4	340	0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	32.8	21.3	31.0	20.7	29.4	20.2	22.9	29.8	22.0	28.9	21.2	27.6	2.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.8	15.8	26.0	19.9	14.9	24.8	19.0	14.1	24.0	68.5	29.7	65.2	28.9	62.1	27.5	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.2	7.0	6.0	DB	-19.4	35.5	3.0	2.3	-21.5	37.1	-23.2	38.5	-24.9	39.7	-27.0	41.4
			WB	-19.6	24.7	3.0	1.2	-21.7	25.5	-23.5	26.2	-25.1	26.8	-27.3	27.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-2.4	-0.1	4.7	10.9	16.4	20.0	21.8	21.3	16.4	10.4	5.1	-0.8		
	DBStd	9.48	5.10	5.49	4.79	3.80	3.48	2.95	2.73	2.91	3.56	4.15	4.48	5.06		
	HDD10.0	1413	384	283	172	34	2	0	0	0	1	47	157	334		
	HDD18.3	3227	642	516	422	225	79	17	4	7	77	248	398	592		
	CDD10.0	1547	0	1	8	60	200	300	367	352	193	58	9	0		
	CDD18.3	320	0	0	0	1	19	67	113	101	19	1	0	0		
	CDH23.3	3364	0	0	1	29	252	688	1127	1022	226	19	0	0		
	CDH26.7	1181	0	0	0	6	53	220	439	406	56	2	0	0		
(14) Wind	WSAvg	2.2	2.4	2.6	2.9	2.6	2.2	1.9	1.8	1.6	1.8	1.8	2.3	2.2		
(15) Precipitation	PrecAvg	546	24	24	27	52	72	77	78	59	49	29	31	24		
	PrecMax	850	69	60	99	127	181	155	145	153	162	122	93	72		
	PrecMin	339	4	7	1	5	22	17	32	5	0	1	3	2		
	PrecStd	118	18	14	21	29	42	36	34	37	43	26	22	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	15.4	20.8	26.7	30.5	33.0	36.0	35.6	30.8	25.7	19.5	12.4		
		MCWB	7.1	9.2	11.3	16.0	18.2	21.8	21.2	21.4	18.9	17.1	13.1	8.1		
	2%	DB	8.3	11.9	17.5	23.1	28.1	30.8	33.0	32.8	28.1	22.5	15.9	9.3		
		MCWB	5.1	7.6	10.3	14.4	17.9	20.9	21.4	21.2	18.2	15.5	11.5	6.9		
	5%	DB	5.7	9.0	14.8	20.7	26.1	29.2	30.9	30.9	26.0	19.6	13.2	7.1		
		MCWB	3.8	6.0	9.2	13.0	17.3	20.0	21.1	20.8	17.6	14.2	10.5	5.1		
	10%	DB	3.9	6.8	12.5	18.3	24.2	27.5	29.0	28.9	23.7	17.1	11.2	5.2		
		MCWB	2.7	4.7	7.7	11.9	16.3	19.3	20.7	20.1	16.6	13.0	9.4	3.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	10.1	12.6	17.0	20.2	23.5	24.1	24.4	21.0	18.3	13.9	9.4		
		MCDB	10.9	14.4	18.5	25.8	27.4	30.5	30.8	31.1	27.9	24.1	17.9	11.0		
	2%	WB	5.5	7.9	10.9	15.2	19.0	22.1	23.0	22.8	19.4	16.4	12.4	7.1		
		MCDB	7.6	10.9	16.7	21.9	26.0	28.9	29.9	29.5	25.5	20.8	15.1	9.2		
	5%	WB	4.1	6.3	9.6	13.8	17.9	21.2	22.1	21.7	18.3	14.8	11.1	5.4		
		MCDB	5.6	8.5	14.3	19.2	24.5	27.5	28.9	28.6	24.3	18.9	12.9	6.7		
	10%	WB	2.7	4.9	8.1	12.5	16.9	20.1	21.2	20.8	17.2	13.4	9.6	3.9		
		MCDB	3.8	6.8	11.8	17.2	22.9	25.8	27.6	27.0	22.5	16.6	11.1	5.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	7.5	9.7	11.3	12.3	12.1	12.6	13.1	11.9	10.4	7.2	6.4		
		MCDBR	9.7	11.5	14.7	15.9	15.6	15.4	16.0	16.7	16.0	14.9	11.6	9.4		
	5% WB	MCWBR	6.8	7.6	8.5	8.1	7.1	6.8	6.4	6.5	7.1	8.3	7.4	6.9		
		MCWBR	8.5	10.2	13.7	14.0	14.1	13.8	13.9	14.5	13.8	13.7	9.7	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.323	0.353	0.392	0.453	0.426	0.424	0.448	0.458	0.413	0.376	0.348	0.321		
	taud		2.313	2.249	2.205	2.065	2.220	2.264	2.206	2.169	2.255	2.338	2.362	2.339		
	Ebn at Noon		754	801	822	801	842	844	817	790	792	767	718	710		
	Edh at Noon		77	101	122	154	137	132	138	138	116	92	74	68		
(44) All-Sky Solar Radiation	RadAvg		1.28	2.05	3.23	4.38	5.61	5.89	5.96	5.42	3.85	2.48	1.33	1.02		
	RadStd		0.17	0.26	0.32	0.40	0.49	0.53	0.50	0.41	0.40	0.29	0.12	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.50	N/A	N/A	+0.81	N/A	N/A	N/A	N/A	+72	+45
(47) Regional (49 neighbors)		+0.62	N/A	N/A	+0.71	N/A	N/A	-138	-183	+72	+16

Nomenclature: See separate page